

POLITECNICO DI MILANO



MATHEMATICAL ENGINEERING, MTM

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EXAM B

Option Calibration and Pricing (HJM multifactor)

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Introduction

This project focuses on the calibration and pricing of options using the multifactorial Heath-Jarrow-Morton (HJM) model, a widely used theoretical framework in computational finance for modeling the evolution of the yield curve or forward prices. In this context, the model is applied to the energy market, specifically to German electricity futures and options on these derivative instruments. The model calibration is based on the given observed Futures' options market prices and involves optimizing the parameters to minimize the discrepancy between modeled and real prices.

1 HJM model with $p=0$, $n=2$

Considering the general HJM model for the swap price $F(t, \tau_1, \tau_2)$, the dynamics under a risk-neutral probability $\mathbb{Q}$, assuming p the number of Brownian Motions and n the number of jump terms, are defined as:

$$F(t, \tau_1, \tau_2) = F(0, \tau_1, \tau_2) \exp \left(\int_0^t A(u, \tau_1, \tau_2) du + \sum_{k=1}^p \int_0^t \Sigma_k(u, \tau_1, \tau_2) dW_k(u) + \sum_{j=1}^n \int_0^t Y_j(u, \tau_1, \tau_2) dJ_j(u) \right)$$

All the following functions are real-valued and continuous representing respectively:

- $A(u, \tau_1, \tau_2)$: the drift,
- $\Sigma_k(u, \tau_1, \tau_2)$: the volatility associated with the Brownian motion $W(u)$,
- $Y_j(u, \tau_1, \tau_2)$: the jump sizes associated at the process $dJ_j(u)$

In our case, with 2 Brownian Motion components ($p=2$) and no jump components ($n=0$), the formula simplifies to:

$$F(t, \tau_1, \tau_2) = F(0, \tau_1, \tau_2) \exp \left(\int_0^t A(u, \tau_1, \tau_2) du + \int_0^t \Sigma_1(u, \tau_1, \tau_2) dW_1(u) + \int_0^t \Sigma_2(u, \tau_1, \tau_2) dW_2(u) \right)$$

The first integral incorporates the drift of the Levy process, while the other two represent the Brownian Motion components. Moreover, we assume $\Sigma_1(t, \tau_1, \tau_2)$ and $\Sigma_2(t, \tau_1, \tau_2)$ to be constant.

To ensure the explicit dynamics of the forward price are correctly modeled under the risk-neutral probability measure $\mathbb{Q}$, the price must follow a martingale process. This condition is always satisfied, for all $0 \leq \tau_1 \leq \tau_2 \leq T$, due to the exponential integrability result outlined in Proposition 6.3 of Benth (2008), which holds in the absence of jumps.

Having met the needed requirements, the admissible parameter range can be simply set as:

$$\begin{cases} \Sigma_1(t, \tau_1, \tau_2) \geq 0 \\ \Sigma_2(t, \tau_1, \tau_2) \geq 0 \end{cases}$$

2 Drift Condition

To maintain arbitrage-free pricing under the risk-neutral measure, we impose a drift condition consistent with the HJM framework. The general drift condition can be expressed as:

$$\begin{aligned} & \int_0^t A(u, \tau_1, \tau_2) du + \frac{1}{2} \sum_{k=1}^p \int_0^t (\Sigma_k(u, \tau_1, \tau_2))^2 du + \sum_{j=1}^n \int_0^t Y_j(u, \tau_1, \tau_2) d\gamma_j(u) + \\ & + \sum_{j=1}^n \int_0^t \int_{\mathbb{R}} \left(e^{Y_j(u, \tau_1, \tau_2)z} - 1 - Y_j(u, \tau_1, \tau_2)z \cdot \mathbb{I}_{\{|z|<1\}} \right) \nu(dz, du) = 0. \end{aligned}$$

In our specific HJM model, where $p = 2$ and $n = 0$, the drift condition simplifies to:

$$\int_0^t A(u, \tau_1, \tau_2) du + \frac{1}{2} \sum_{k=1}^2 \int_0^t (\Sigma_k(u, \tau_1, \tau_2))^2 du = 0. \quad (1)$$

Having constant volatility we can modify the previous condition as:

$$\int_0^t A(u, \tau_1, \tau_2) du = -\frac{1}{2} \left(\Sigma_1(t, \tau_1, \tau_2)^2 + \Sigma_2(t, \tau_1, \tau_2)^2 \right) \cdot t$$

This drift condition ensures that the explicit dynamics of the forward are set under the risk-neutral probability $\mathbb{Q}$.

Option Pricing Using Volatilities

To calculate the prices of European options on futures, we employed the Black-76 formula, an adaptation of the Black-Scholes model for futures. The formula is given by:

$$C = DF \cdot (F_0 \cdot N(d_1) - K \cdot N(d_2)), \quad (1)$$

where DF represents the discount factor, F_0 is the forward price of the future, K is the strike price, and $N(\cdot)$ is the cumulative distribution function of the standard normal distribution. The parameters d_1 and d_2 are defined as:

$$d_1 = \frac{\ln(F_0/K) + \frac{1}{2} \cdot \sigma^2}{\sigma} \quad d_2 = d_1 - \sigma. \quad (2)$$

Option prices were derived using market-implied volatilities. The Black-76 formula is particularly suitable for futures as it accounts for the absence of intermediate cash flows and evaluates the payoff at maturity.

The following figure illustrates the market option prices and market implied volatilities.

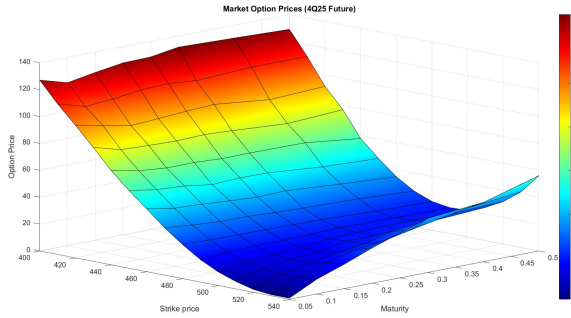


Figure 1: Option Prices

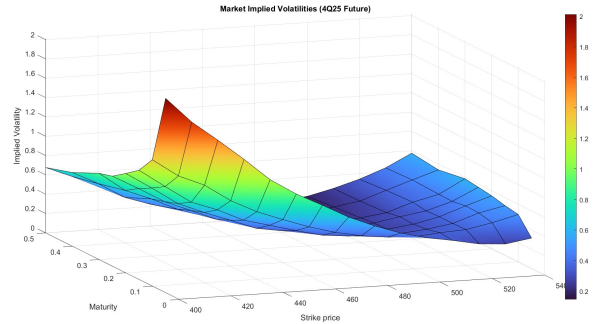


Figure 2: Implied Volatilities

The 3D surface plots of market-implied volatilities and option prices for the 4Q25 German power future highlight the well-known "volatility smile". These surfaces reveal important relationships between implied volatilities, option prices, strike prices, and maturities.

Volatility Smile Across Strike Prices

Implied volatilities are lowest near at-the-money (ATM) strikes and increase significantly for deep in-the-money (ITM) and out-of-the-money (OTM) options, forming the characteristic "smile". This pattern reflects the market's adjustment for the heightened risk of extreme price movements at these strike levels.

Dependence on Maturities

At shorter maturities, the smile is more pronounced, with steeper increases in implied volatilities at extreme strikes. This indicates heightened sensitivity to strike prices over short time horizons. For longer maturities, the smile flattens, reflecting diminishing market expectations for extreme price movements due to mean-reverting tendencies in power markets.

Option Prices Derived from Volatilities

Lower strike prices correspond to higher option prices, driven by the higher intrinsic value of ITM options and increased implied volatilities. Conversely, higher strike prices yield lower option prices, as these OTM options are less likely to be exercised profitably, although they retain higher implied volatilities due to market perceptions of risk.

3 Option Calibration and Pricing (HJM multifactor)

Model and Assumptions

The pricing dynamics for the underlying futures are described by the equation:

$$F(t, \tau) = F(0, \tau) \cdot \exp \left(\int_0^t A(u, \tau) du + \sum_{k=1}^2 \int_0^t \Sigma_k dW_k(u) \right), \quad (3)$$

where:

- $F(t, \tau)$ denotes the forward price of the futures contract at time t ,
- $A(u, \tau)$ is the drift term,
- Σ_k are the volatility parameters for the independent Brownian motions $W_k(u)$.

The following key assumptions underpin the model:

1. **Constant Volatility:** The parameters Σ_1 and Σ_2 are fixed over time.
2. **No Jumps:** Jump processes are excluded to simplify the model's structure.
3. **Martingale Property:** The model ensures risk-neutral pricing, precluding arbitrage opportunities.

The primary goal of calibration was to optimize the parameters Σ_1 and Σ_2 to align the model-generated call option prices with observed market data.

Calibration Methodology

The DATA_DEEEX file contains all liquid maturities for the German power futures and the call option prices (implied volatility quotes) on the 4Q25 futures as of the 4th of November 2024. The calibration was performed by minimizing the discrepancy between theoretical call option prices and observed market prices. The process involved the following steps:

Objective Function

The calibration was driven by minimizing the mean squared error (MSE) between model-derived and market-observed prices:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (P_{\text{model},i} - P_{\text{market},i})^2, \quad (4)$$

where $P_{\text{model},i}$ represents the model price of the i -th option, $P_{\text{market},i}$ is the corresponding market price, and N is the number of observations.

Pricing Formula

The theoretical prices of call options were computed using the Black-76 formula, adapted for futures:

$$C(0, T, K) = e^{-rT} [F(0, T)\Phi(d_1) - K\Phi(d_2)], \quad (5)$$

where:

$$d_1 = \frac{\ln(F(0, T)/K) + \frac{1}{2}\hat{\sigma}^2}{\hat{\sigma}},$$
$$d_2 = d_1 - \hat{\sigma}.$$

The aggregated volatility $\hat{\sigma}$ is defined as:

$$\hat{\sigma} = \sqrt{\Sigma_1^2 T + \Sigma_2^2 T}, \quad (6)$$

with T representing the option's time to maturity.

Optimization

The calibration leveraged numerical optimization techniques to determine Σ_1 and Σ_2 . The problem was formulated as:

$$\min_{\Sigma_1, \Sigma_2} \text{MSE}, \quad \text{subject to} \quad \Sigma_1 \geq 0, \quad \Sigma_2 \geq 0. \quad (7)$$

The positivity constraints ensured that the parameters were physically and mathematically meaningful, reflecting valid volatility measures. The aggregated volatility $\hat{\sigma}$ was calculated for each maturity T , integrating the effects of the two independent Brownian motion components.

Evaluation of the Calibration Process

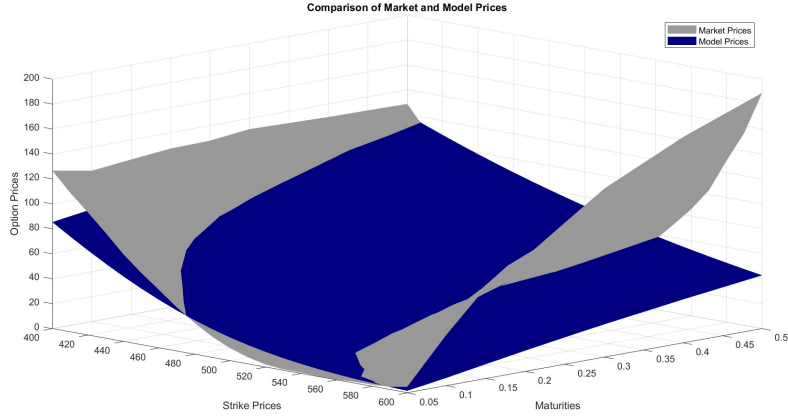


Figure 3: Calibration before filtering

The initial calibration model exhibited a high Mean Squared Error (MSE) of approximately 1000, particularly for options with extreme strike prices near the lower end of 400 and the upper end of 600, as showed in Figure: 3. The disparity between model and market prices in these regions, as seen in the analysis, highlighted the limitations of including these extreme strikes in the calibration.

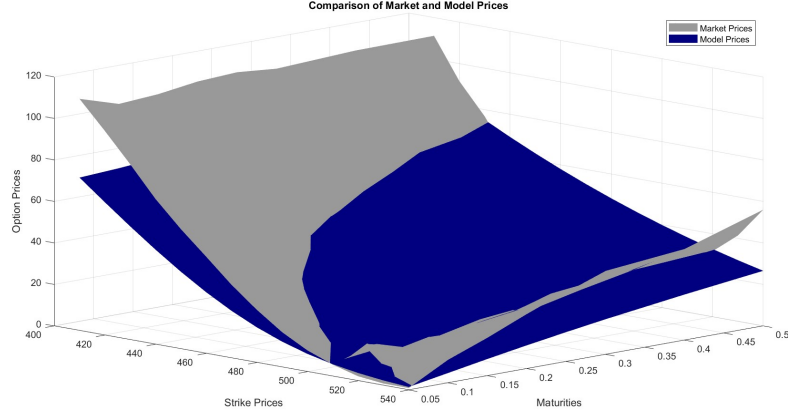
Adjusting the Strike Range to 410–540

To address these issues, the calibration was refined to focus on a strike price range of [410–540], excluding the problematic extremes.

By narrowing the range to the central portion of the strike distribution, the calibration prioritizes the most reliable and relevant data, avoiding noise and extreme sensitivities.

Benefits of the Refinement

The exclusion of extreme strike prices significantly reduced the MSE to 231.963, leading to a better alignment between the model and market dynamics. The refined calibration ensured that the model captured the most meaningful aspects of the volatility surface, focusing on regions with high data reliability and calibration relevance, moreover avoiding overfitting.



The calibration approach successfully optimized the volatility parameters, aligning the model-generated prices with the observed market data. Despite relying on significant simplifications, such as constant volatility, the model effectively captures market dynamics and provides a robust foundation for pricing call options in energy markets. This solid baseline highlights the potential for further improvements. Introducing time-dependent volatilities could enhance the model's ability to generalize, enabling better replication of complex market behaviors while preserving computational efficiency.

The calibrated value for the constant Σ_1 and Σ_2 are:

$$\begin{cases} \Sigma_1 = 0.2458 \\ \Sigma_2 = 0.2492 \end{cases}$$

For the subsequent sections, all computations were performed using the filtered set of options.

4 Calibration of the model on the 4Q25 German option prices by considering a generic time dependent $\Sigma_1(t, \tau_1, \tau_2)$

Introducing a time-dependent volatility in the model, would significantly enhances the model's ability to capture changes in short and long-term volatility. Volatility is rarely constant; it fluctuates over time in response to economic events, political developments, or market-specific factors such as supply-demand imbalances. A time-dependent volatility allows the model to account for these variations more effectively, enabling it to adapt to dynamic market conditions and better align with observed option prices across different maturities. This adaptability ensures the model remains robust and realistic in the face of ever-changing market environments. Therefore, the volatility function of the first Brownian motion, $\Sigma_1(t, \tau_1, \tau_2)$, was introduced as a time-dependent component. Conversely, the second volatility Σ_2 was held constant to reduce complexity and focus the analysis on Σ_1 .

The calibration centered around the cumulative impact of Σ_1 , which is expressed as the integral of its square over the time-to-maturity:

$$\int_0^T \Sigma_1^2(u, \tau_1, \tau_2) du. \quad (8)$$

This integral encapsulates the volatility's contribution to the option price, and modeling it directly avoids the need to specify $\Sigma_1(t, \tau_1, \tau_2)$ in detail. By calibrating this integral, the expectation is that the model achieves a closer alignment with market data without compromising computational efficiency.

To ensure logical consistency, the integral was divided into increments over successive maturity intervals:

$$\int_{T_{n-1}}^{T_n} \Sigma_1^2(u, \tau_1, \tau_2) du, \quad (9)$$

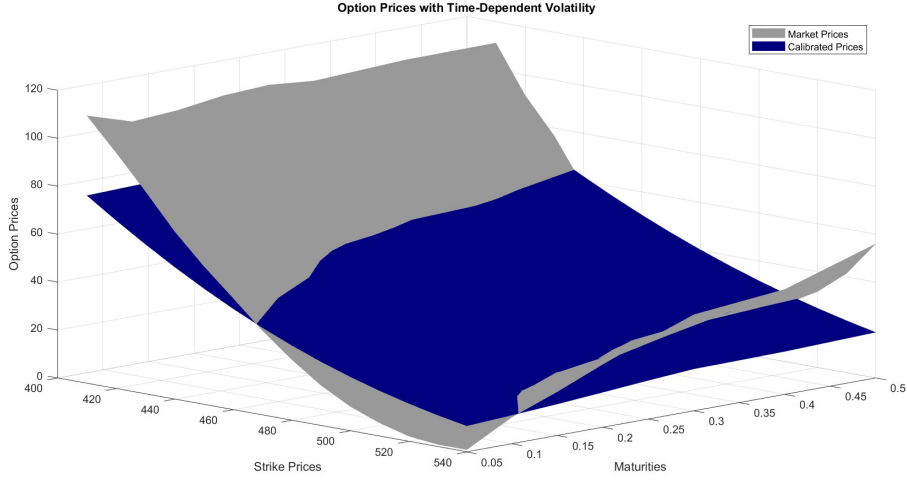
where T_n represents the set of maturities under consideration, with $T_0 = 0$. This structure guarantees a monotonic increase in the cumulative integral, as volatility is inherently non-negative.

For Σ_2 , its constant nature allows for straightforward integration:

$$\int_0^T \Sigma_2^2(u, \tau_1, \tau_2) du = \Sigma_2^2 \cdot T. \quad (10)$$

This simplification isolates the calibration process to $\Sigma_1(u, \tau_1, \tau_2)$, enabling a focused and efficient optimization.

This approach provides the model with significant adaptability, ensuring that it can accommodate a variety of market scenarios without being constrained by rigid assumptions. The emphasis on the integral form of $\Sigma_1^2(u, \tau_1, \tau_2)$ allows the model to capture complex dynamics and better reflect the observed option prices while maintaining logical and mathematical consistency.



It's quite interesting the comparison between the two calibration approaches of point III and IV. The MSE value decreased from 231.9636 to 198.5886 both models achieve a reasonable representation of market option prices, with certain distinctions emerging in specific regions.

The constant volatility model (point III) performs well in capturing the general trends of the market prices, particularly for at-the-money (ATM) options where the volatility is more stable. However, it shows some limitations in regions with extreme strike prices, such as deep in-the-money (ITM) or out-of-the-money (OTM) options. These discrepancies arise from the inability of the constant volatility assumption to adapt to the more complex behavior of implied volatilities in these areas.

On the other hand, the time-dependent volatility model demonstrates a slightly closer alignment with market prices, particularly in the regions where implied volatilities exhibit greater variability. Its flexibility allows it to better capture the volatility smile and adjust to the varying dynamics across different maturities. Despite this, the improvements in accuracy are relatively subtle, suggesting that the constant volatility model remains competitive in many cases.

In the central region of the strike price-maturity surface, corresponding to ATM options, from the graphs, the two models produce very similar results. This is expected, as the volatility in these regions tends to be less sensitive to extreme market conditions.

The calibrated value for $\Sigma_2(t, t_1, t_2)$ is: 0.0470.

The time-dependent volatility model Σ_1 introduces additional flexibility compared to the constant volatility approach. However, its effectiveness in predicting option prices remains limited due to structural and practical constraints. One key limitation lies in the sparse time grid of tenor points, which prevents the model from fully capturing the dynamic behavior of Σ_1 . This grid, with only a few points, restricts the precision of the cumulative volatility integral, reducing the potential advantage of a time-dependent framework.

Maturity	Σ_1 increments	Σ_2 increments
0.05	0.023469	0.00010841
0.1	0.0020567	0.00010841
0.15	0.002456	0.00010841
0.2	0.0030489	0.00010841
0.25	0.0032964	0.00010841
0.3	0.0029756	0.00010841
0.4	0.00097001	0.00021682
0.5	0.0025605	0.00021682

Table 1: Sigma hat increments from Σ_2 and Σ_1 for different maturities.

The second volatility parameter Σ_2 , as we can see in table 1, contributes negligibly to the overall model volatility (σ_{hat}). Its impact is consistently found to be an order of magnitude smaller than that of Σ_1 , making its role almost irrelevant in determining the total volatility. As a result, the pricing dynamics are driven primarily by Σ_1 (equation: 9), while Σ_2 has little influence on the model’s performance.

Therefore, we can conclude that the integral of Σ_1 over the time horizon plays the most significant role in the model. This integral effectively encapsulates the temporal evolution of volatility and is the dominant term shaping the option price. Consequently, while Σ_2 and other factors might theoretically contribute, the integral of Σ_1 is the critical component for accurate pricing.

5 Calibration of the model on the 4Q25 German option prices by considering a generic time dependent $\Sigma_1(t, \tau_1, \tau_2)$ and $\Sigma_2(t, \tau_1, \tau_2)$

To investigate whether further improvements in accuracy can be achieved, we extended the calibration process to include two fully generic time-dependent volatility functions, $\Sigma_1(t, \tau_1, \tau_2)$ and $\Sigma_2(t, \tau_1, \tau_2)$. In this approach, the model was designed to adjust both volatility functions by directly modeling their respective integral increments over time.



Figure 4: One time dependent volatility

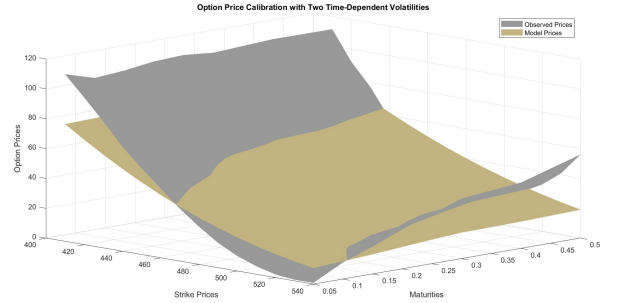


Figure 5: Two time dependent volatilities

The Mean Squared Error (MSE) from the calibration using a single time-dependent volatility is nearly identical to that obtained with the generalized model highlights a key insight. This outcome can be explained by the specific characteristics of the dataset and the calibration process.

Characteristics of the Market Data

The implied volatility surface observed in the market options does not appear to exhibit sufficient complexity to necessitate the use of a more generalized model. A single time-dependent volatility term seems to effectively capture the overall dynamics of the market prices across maturities and strike prices. This suggests that the variability in the market data is relatively smooth and well-approximated by the simpler model (point IV).

The generalized model introduces additional degrees of freedom by allowing for a more flexible volatility structure. However, the calibration process is constrained by the observed data. If the dataset is not sufficiently granular or does not exhibit pronounced non-linearities, the optimization algorithm will naturally converge to solutions similar to those obtained with the simpler time-dependent model. In this case, the additional flexibility

remains unused because the market data does not demand it.

Implications

The similarity in MSE values between the two approaches indicates that the simpler time-dependent volatility model is sufficient for this dataset. The generalized model does not significantly improve accuracy, as the observed market dynamics are already well-represented by the simpler approach. This result emphasizes the importance of aligning model complexity with the actual characteristics of the dataset. While the generalized model offers theoretical advantages, these are not realized in practice unless the data demands additional flexibility.

6 Pricing a down and in call option with the model calibrated in previous sections

To address the pricing of a *down-and-in call option* with a barrier $L = 450$, strike $K = 500$, and maturity of 6 months, three different models calibrated in prior steps were applied: *constant volatility*, *single time-dependent volatility*, and *dual time-dependent volatility*. Below, the methodologies and key steps used for pricing are detailed.

Pricing Methodology

Monte Carlo Simulations The Monte Carlo approach was used consistently across all models. The steps involved:

1. **Path Simulation:** Simulating trajectories of the underlying F_t for the given maturity T under each model

$$F(T_n, \tau_1, \tau_2) = F(T_{n-1}, \tau_1, \tau_2) \cdot \exp \left\{ -\frac{1}{2} \int_{T_{n-1}}^{T_n} \Sigma_1^2(u, \tau_1, \tau_2) + \Sigma_2^2(u, \tau_1, \tau_2) du + \sum_{k=1}^2 \int_{T_{n-1}}^{T_n} \Sigma_k^2(u, \tau_1, \tau_2) dW_k(u) \right\}$$

where Σ_1 and Σ_2 represent the volatility components.

2. **Barrier Crossing:** Determining if the trajectory breached the barrier $L = 450$ at any point during its evolution.
3. **Payoff Calculation:** For paths that breached the barrier, the payoff was computed as:

$$C_{DI} = \mathbb{E} \left[\max(F_T - K, 0) \cdot \mathbf{1}_{\{\text{barrier crossed}\}} \right] \cdot DF,$$

where F_T is the terminal value of the underlying, $K = 500$ is the strike, and DF is the discount factor.

Closed-Form Formula for Constant Volatility In the case of constant volatility, a closed-form solution exists, which is referenced in Theorem 18.8 of Arbitrage Theory in Continuous Time by Tomas Bjork.

$$C_{DI} = \left(\frac{L}{F_0} \right)^{\frac{2\hat{r}}{\sigma^2}} \cdot C_{\text{vanilla}},$$

where:

- $\hat{r} = r - \frac{\sigma^2}{2}$ is the adjusted risk-free rate,
- $\sigma^2 = \Sigma_1^2 + \Sigma_2^2$ is the total volatility,
- C_{vanilla} is the price of a vanilla call option with adjusted parameters:

$$C_{\text{vanilla}} = DF \cdot \left(\frac{L^2}{F_0} \cdot N(d_1) - K \cdot N(d_2) \right),$$

with:

$$d_1 = \frac{\ln \left(\frac{L^2}{F_0 \cdot K} \right) + \frac{1}{2} \sigma^2}{\sigma}, \quad d_2 = d_1 - \sigma.$$

This formula serves as a benchmark for the constant volatility model and is derived from Black-76.

Implementation

- **Constant Volatility Model:**

$$\int_{T_{n-1}}^{T_n} \Sigma_1^2(u, \tau_1, \tau_2) + \Sigma_2^2(u, \tau_1, \tau_2) du = (\Sigma_1^2 + \Sigma_2^2) \cdot (T_n - T_{n-1}).$$

Paths were simulated using this formula, and the option price was derived through Monte Carlo.

- **Single Time-Dependent Volatility Model:** The time-dependent volatility included a constant component (Σ_2) and an incremental component (Σ_1) that varied with time:

$$\int_{T_{n-1}}^{T_n} \Sigma_1^2(u, \tau_1, \tau_2) du + \Sigma_2^2 \cdot (T_n - T_{n-1}).$$

Paths were simulated accordingly, and the payoff was computed similarly to the constant volatility case.

- **Dual Time-Dependent Volatility Model:** Both volatility components (Σ_1 and Σ_2) were allowed to vary with time, and their contributions were computed individually for each time interval:

$$\int_{T_{n-1}}^{T_n} \Sigma_1^2(u, \tau_1, \tau_2) + \Sigma_2^2(u, \tau_1, \tau_2) du.$$

Monte Carlo simulations were conducted, and the down-and-in option price was obtained, below are reported the results:

Model	Const. Vol.	Single T-Dep. Vol.	Dual T-Dep. Vol.
Down-and-In Price	8.1731	1.4221	1.4189
Closed Formula Price: 14.6670			

The pricing of the down-and-in call option with a barrier $L = 450$, a strike $K = 500$, and a maturity of 6 months highlights significant differences among the three models used for pricing: the constant volatility model, the single time-dependent volatility model, and the dual time-dependent volatility model.

Constant Volatility Model

The constant volatility model produced a Monte Carlo price of **8.1731**, which is significantly lower than the closed-form price of **14.6670**. This discrepancy arises because the Monte Carlo simulation monitors the barrier only at discrete market maturities, whereas the closed-form formula assumes continuous monitoring. Continuous monitoring increases the likelihood of the barrier being breached, resulting in a higher theoretical value for the option. Despite this gap, the closed-form solution serves as a valuable benchmark to validate the Monte Carlo implementation for the constant volatility model.

Single Time-Dependent Volatility Model

The single time-dependent volatility model incorporates a time-varying component Σ_1 while keeping Σ_2 constant. This added flexibility increases the complexity of path simulations. However, the results reveal that the variability of Σ_1 lowers the probability of the barrier being crossed, which reduces the option's price to **1.4221**. This outcome indicates that Σ_1 's variability might hinder the option's ability to breach the barrier, especially under the discrete monitoring framework.

Dual Time-Dependent Volatility Model

The dual time-dependent volatility model allows both Σ_1 and Σ_2 to vary with time. However, its resulting price of **1.4189** is nearly identical to the single time-dependent model's price. This similarity suggests that the additional variability introduced by Σ_2 has a negligible impact on the overall barrier dynamics.

Observations on Model Performance

The availability of a closed-form solution offers a robust benchmark, enhancing confidence in its accuracy. The time-dependent volatility models, while theoretically more flexible, yield significantly lower prices that are less reflective of market dynamics. This discrepancy may stem from the limitations of discrete monitoring and the marginal impact of Σ_2 in these scenarios.

Relevance of the Closed-Form Solution

The closed-form solution for the constant volatility model within the constant volatility framework provides the highest value among the models and serves as an essential benchmark. The gap between the Monte Carlo price

and the closed-form solution underscores the importance of monitoring frequency. Increasing the monitoring frequency in simulations would likely bring the Monte Carlo price closer to the theoretical value, as the higher resolution captures more realistic barrier crossings.

Concluding Insights

The near-identical results of the single and dual time-dependent volatility models suggest that the additional complexity introduced by a time-varying Σ_2 does not meaningfully improve pricing accuracy in our case. Furthermore, the negligible contribution of Σ_2 confirms that the **integral of Σ_1** is the dominant factor in determining the effective volatility and barrier dynamics. This supports the conclusion that the constant volatility model, with its closed-form benchmark and simplicity, remains the most reliable approach under the given dataset and assumptions.